



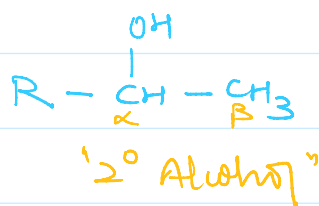
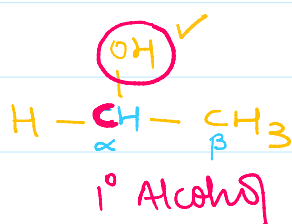
Alkyl & Aryl Halide.

- IUPAC Nomenclature
- Preparation
- Properties

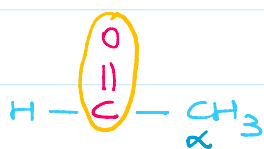
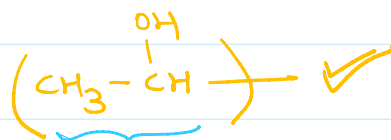
Haloform Test :-

1° & 2° alcohol having β -methyl group ; aldehyde & ketones having α -methyl group on treatment with excess of X_2 and alkali produces haloform & carboxylate salt.

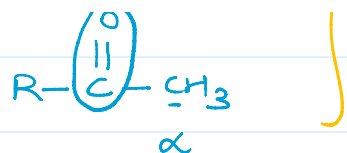
- $CHCl_3$ = Chloroform
 - $CHBr_3$ = Bromoform
 - CHI_3 = Iodoform
- ↓
"yellow solid"



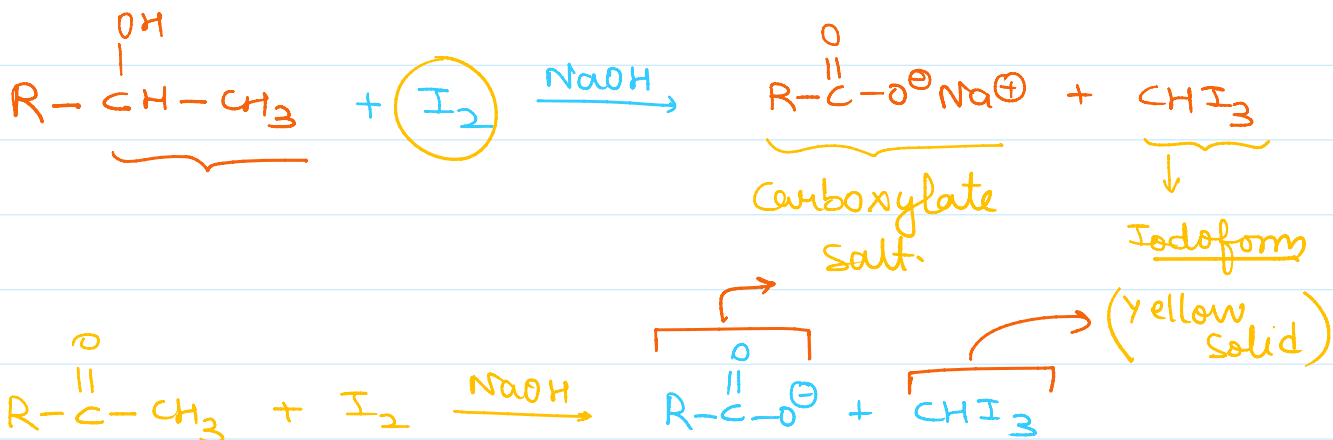
1° & 2° alcohols having β -methyl Group



Ald. / Ketones having α -methyl group.

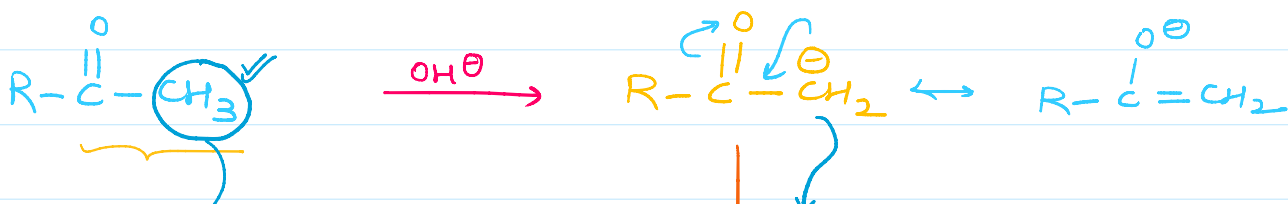
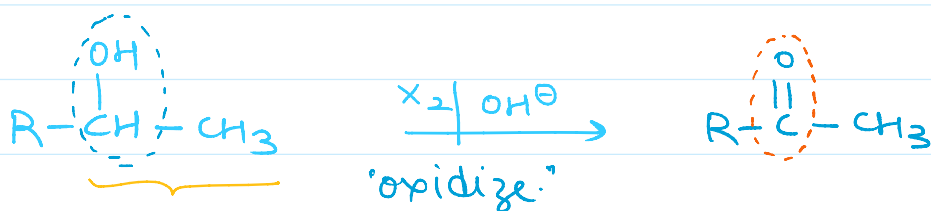
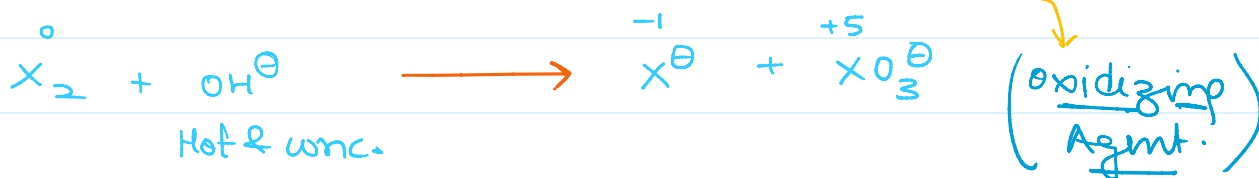
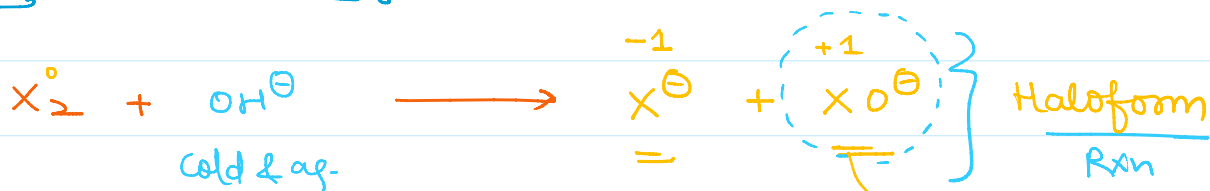


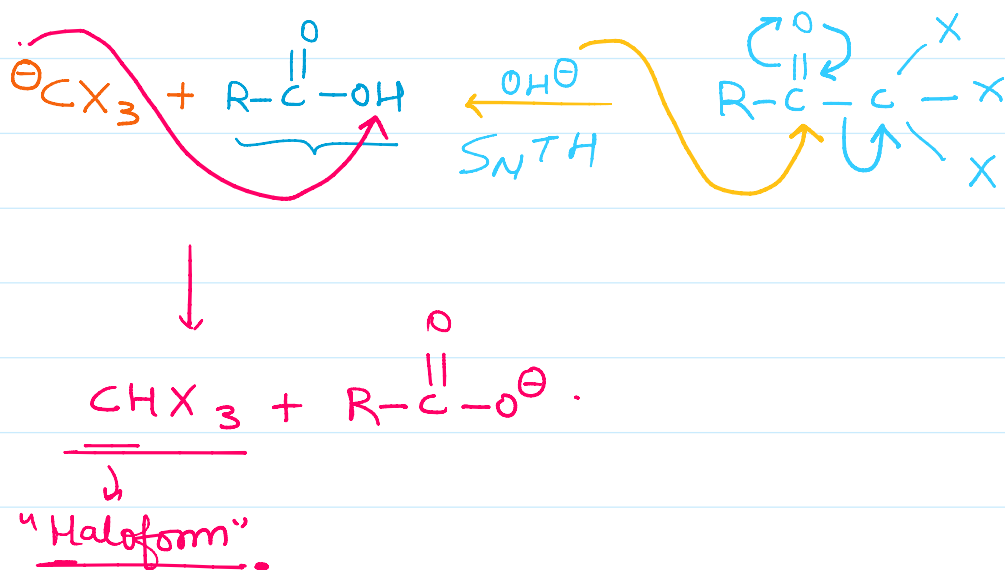
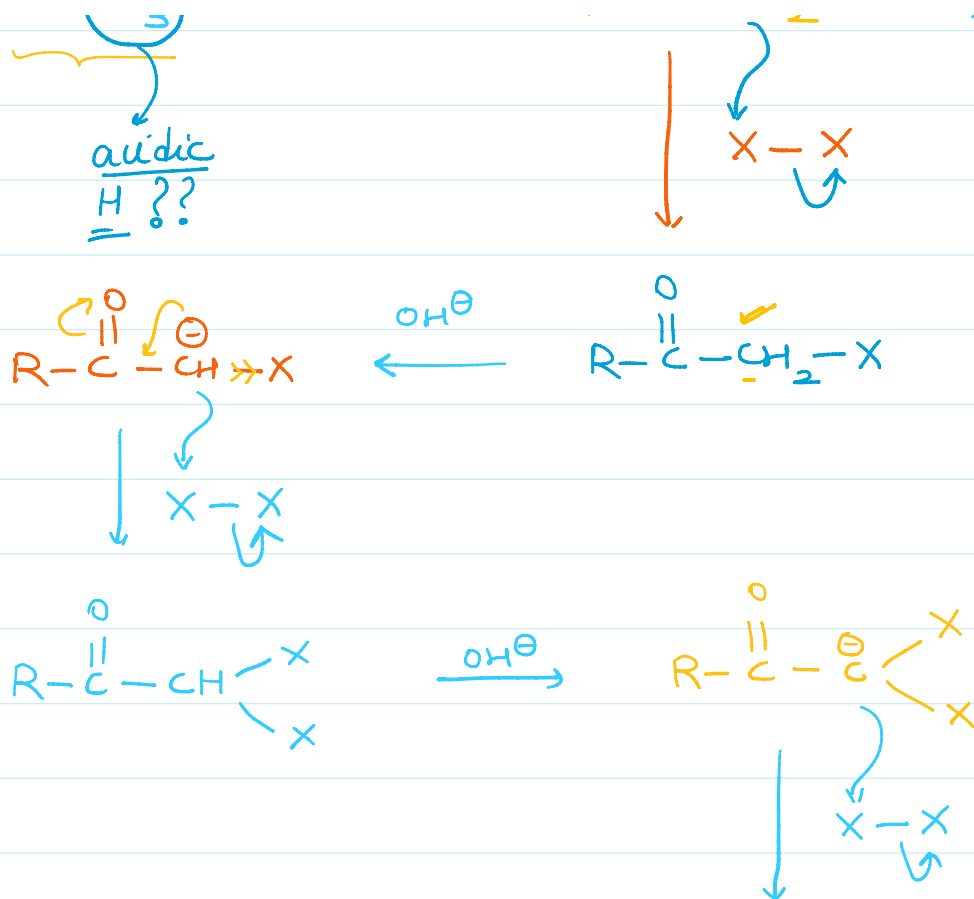
α -methyl group.



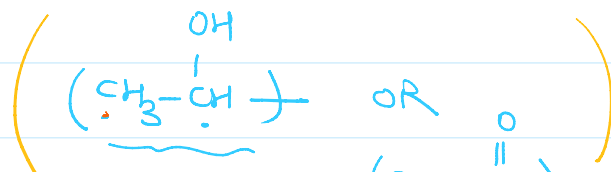
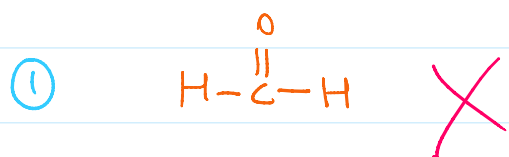
Mechanism:-

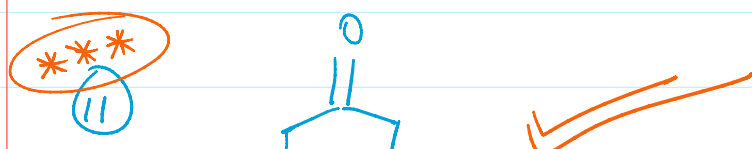
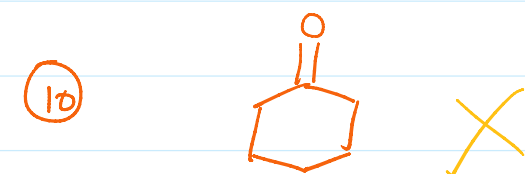
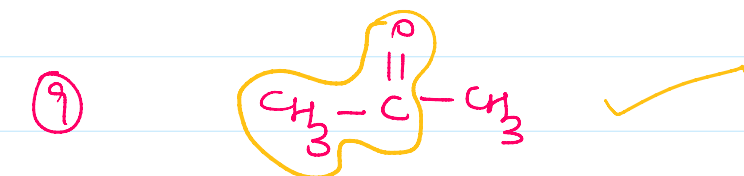
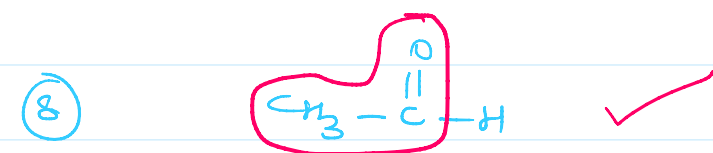
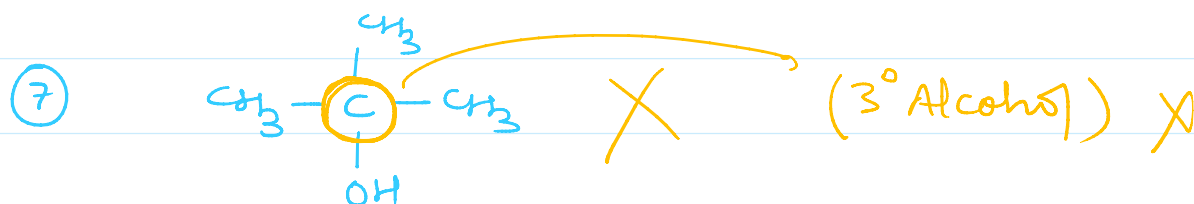
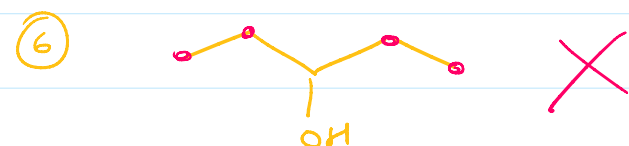
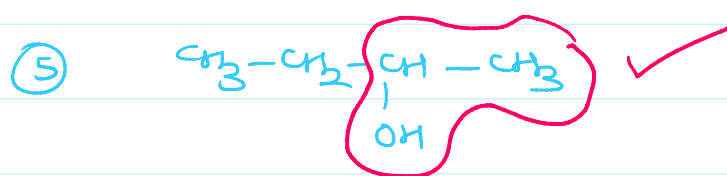
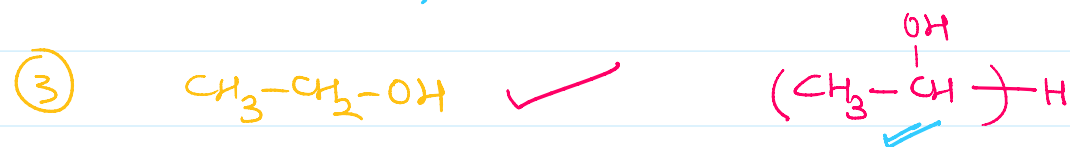
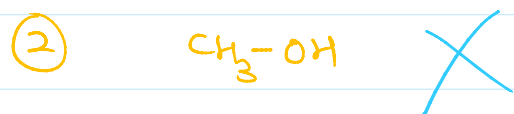
Group 17 Halogen family:

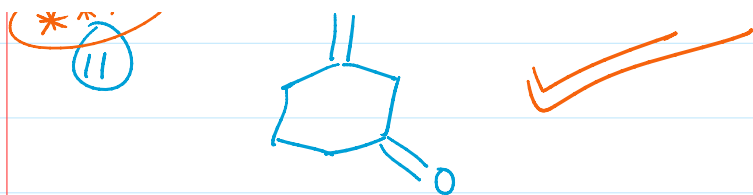




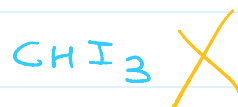
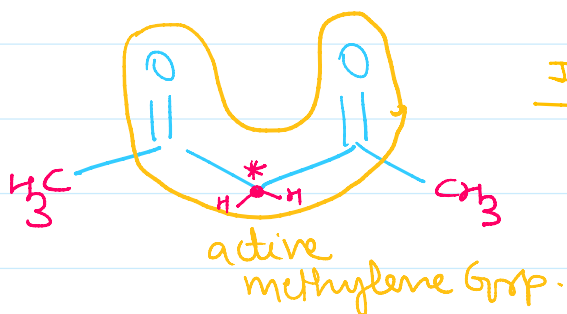
Q. Which of the following compounds give +ve Haloform Test?



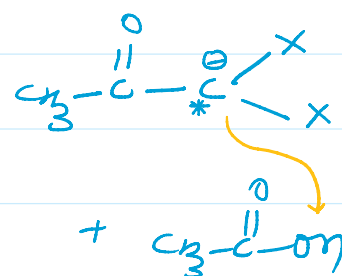
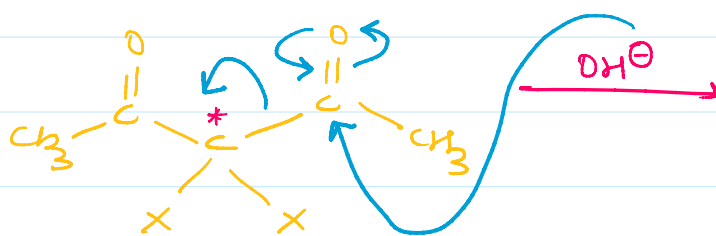
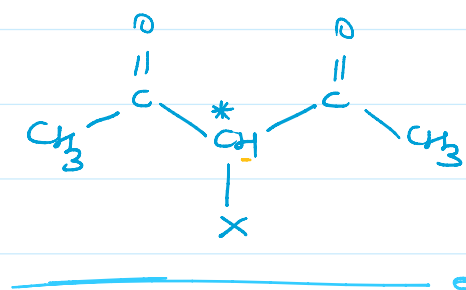
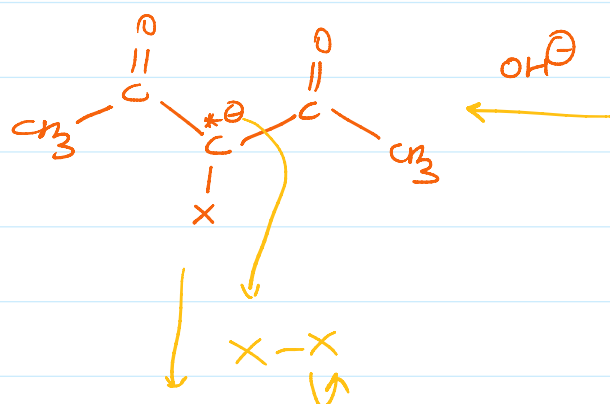
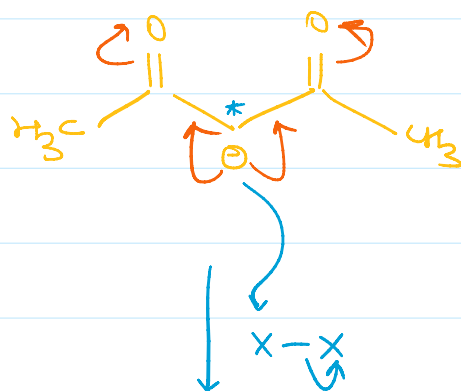
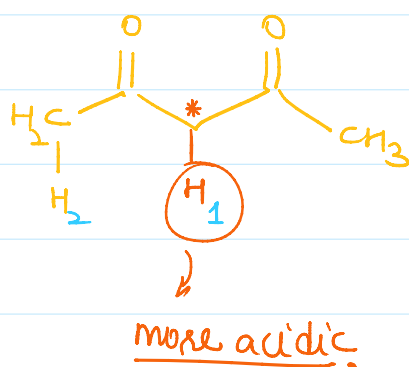


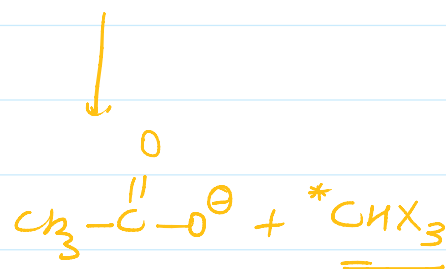
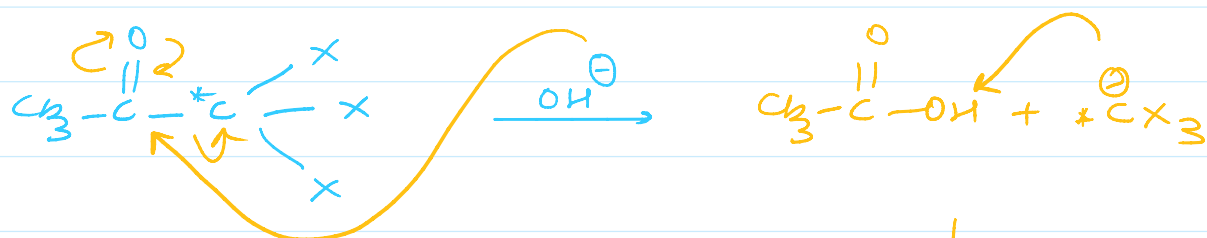
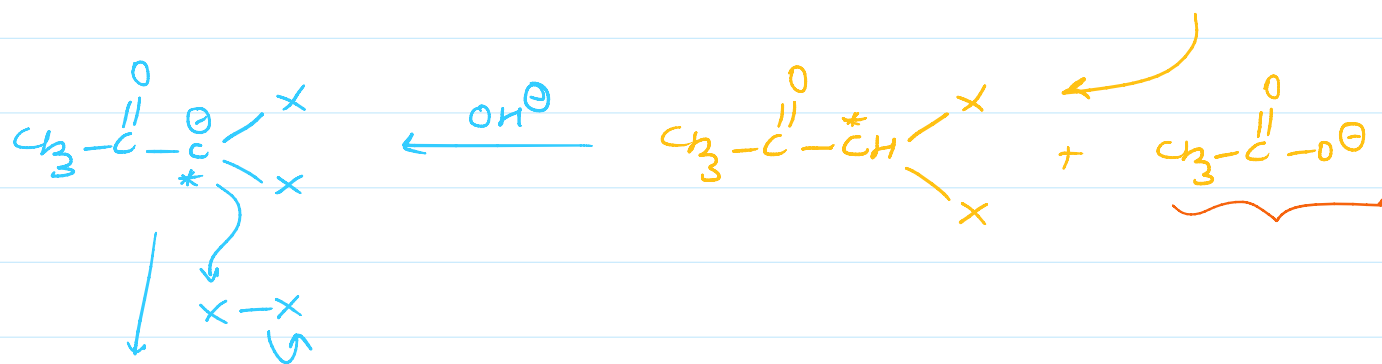


Compounds having Active methylene Groups:-

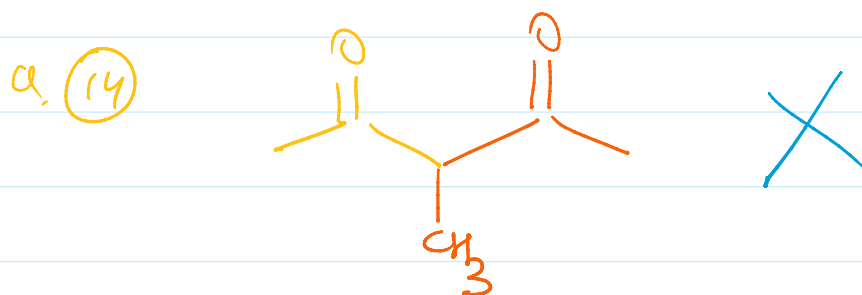
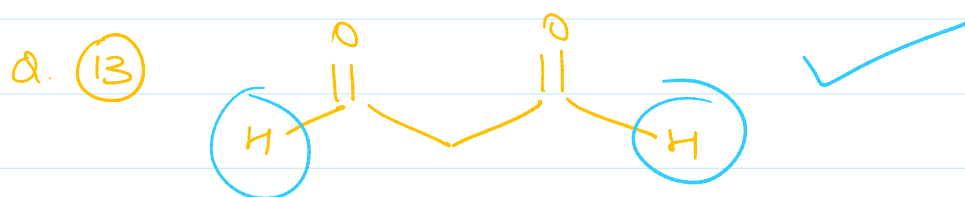
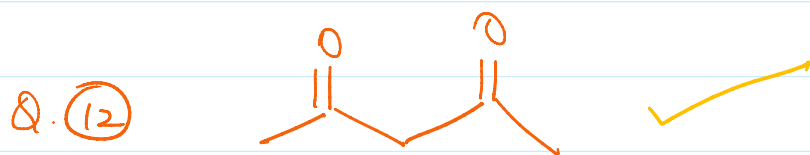
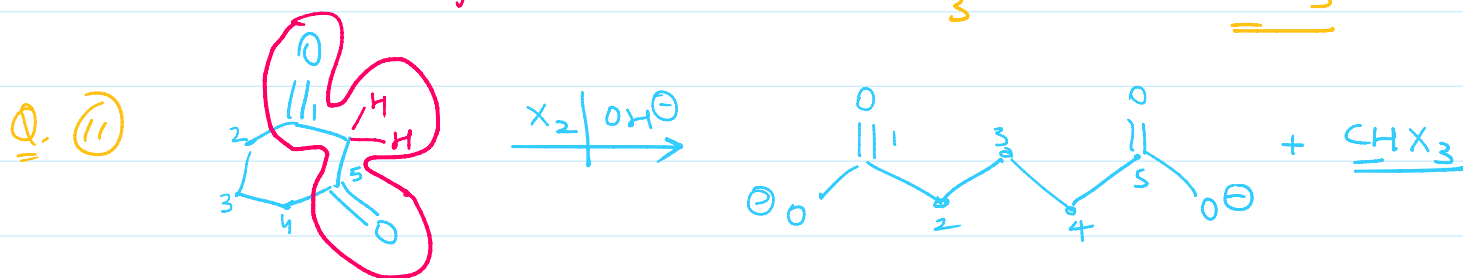


Mech:

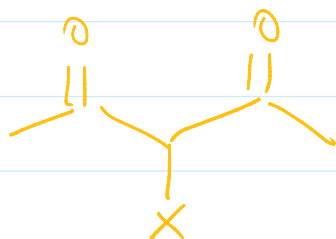




active methylene Grp.

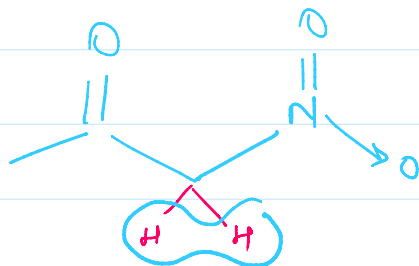


Q. (15)

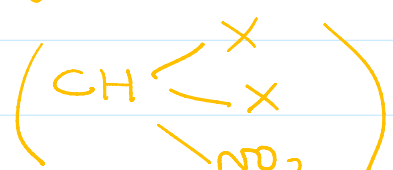
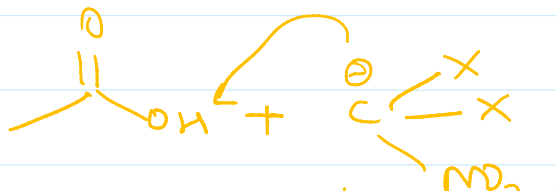
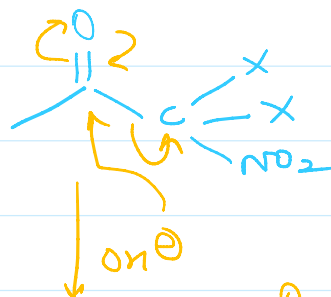
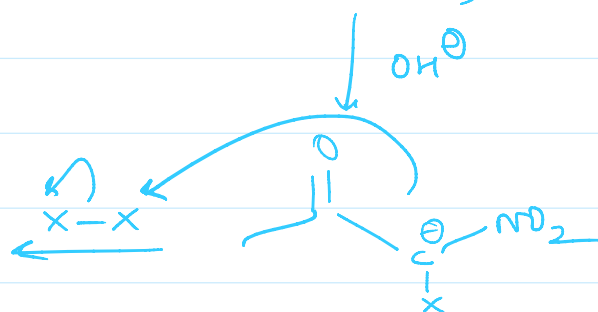
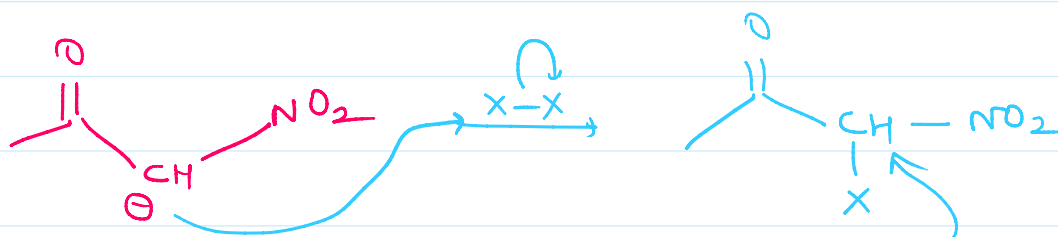
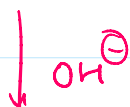


✓

Q. (16)

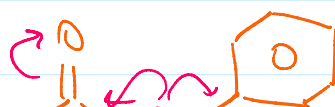


✗



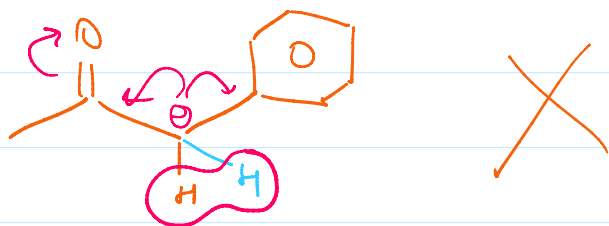
CHX₃

Q. (17)

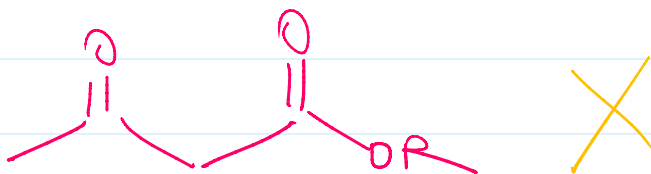


✓

Q. (17)



Q. (18)

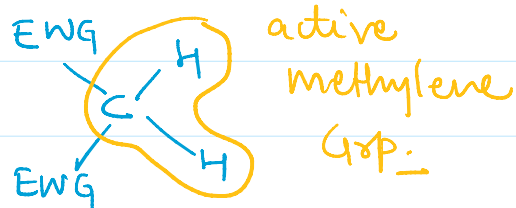


Haloforn Test ✓

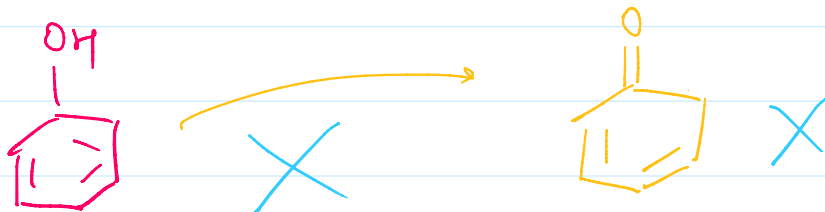
EWG = $-\overset{\text{O}}{\parallel}{\text{C}}-$ only

Haloforn Test ✗

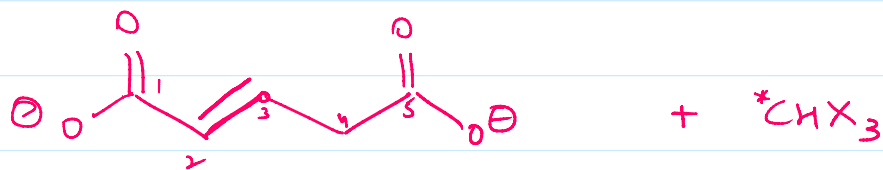
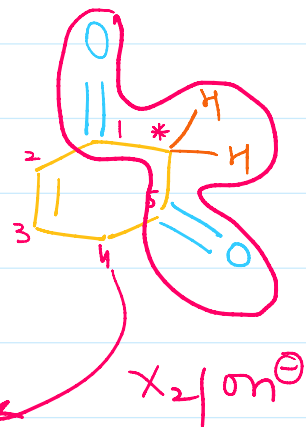
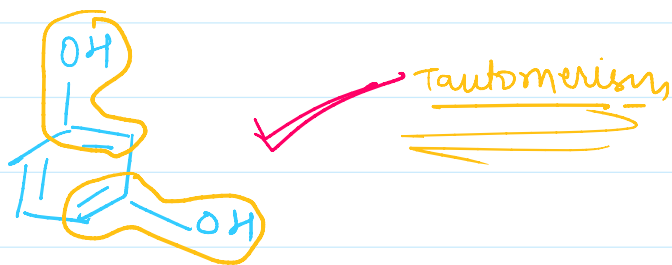
EWG = $-\text{NO}_2$ | $-\text{CN}$ | O
 $-\overset{\text{O}}{\parallel}{\text{C}}-\text{OR}$ etc.



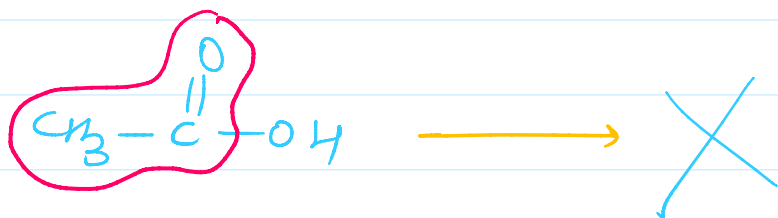
Q. (19)



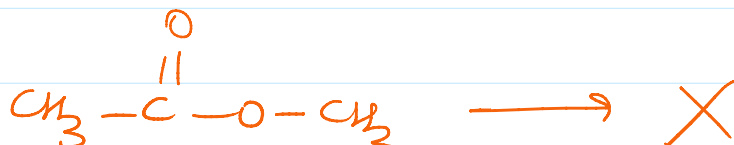
Q. (20)



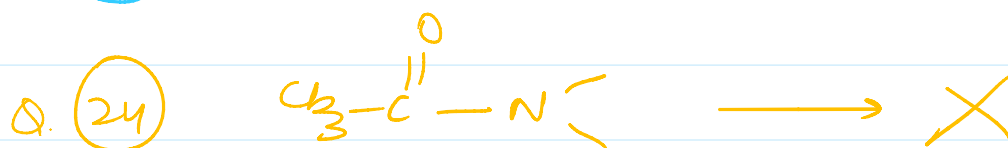
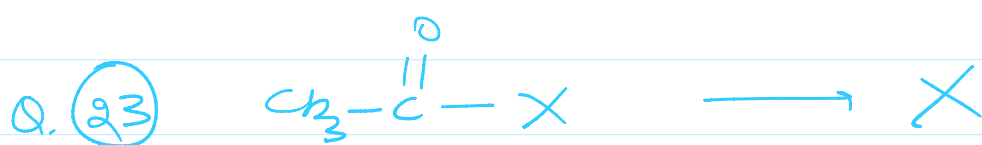
Q. (21)



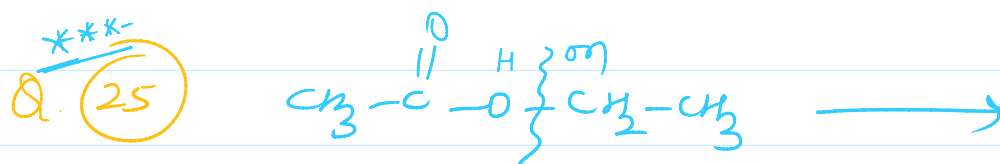
Q. (22)



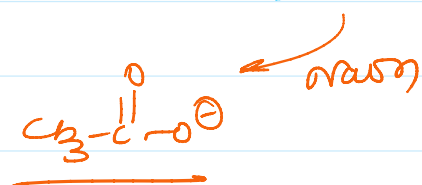
Acid & acid



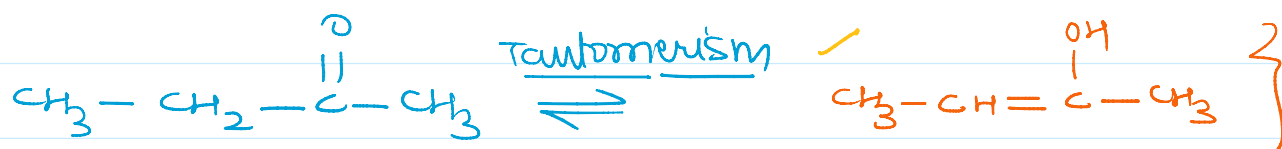
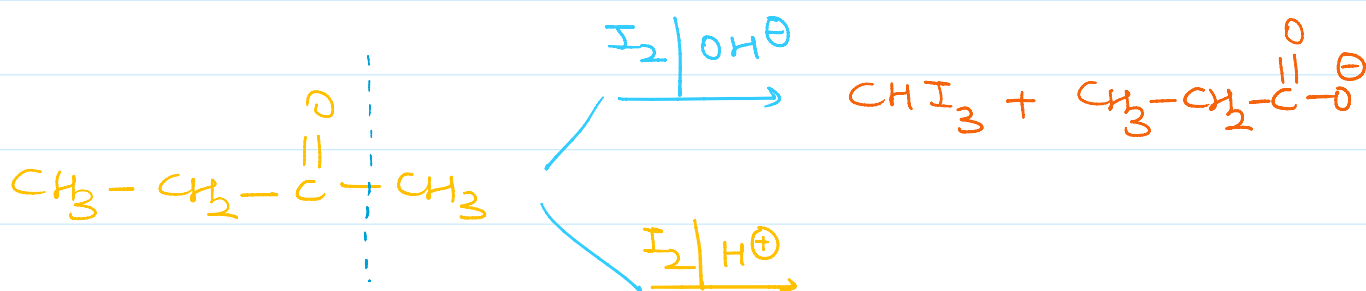
acid derivatives
do not show
Haloform
Test.

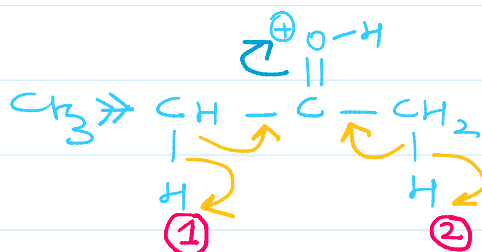
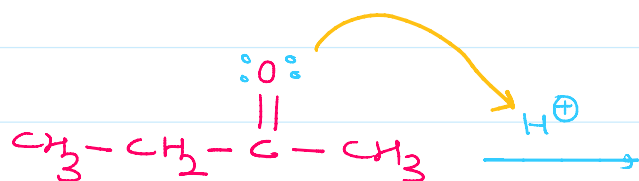
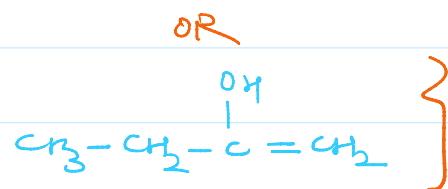
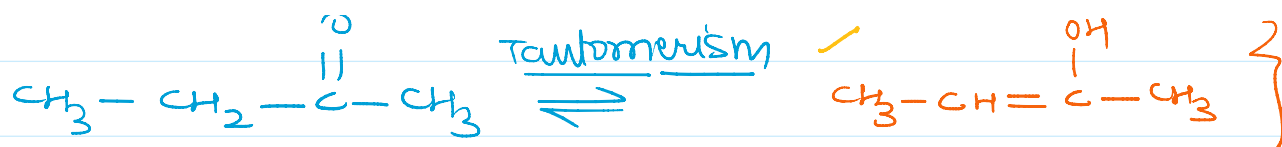


↓ aq. NaOH

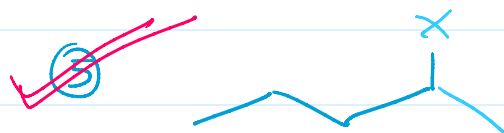
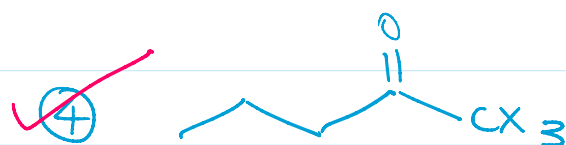
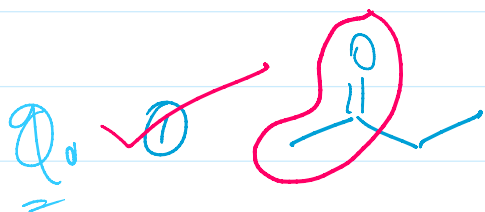
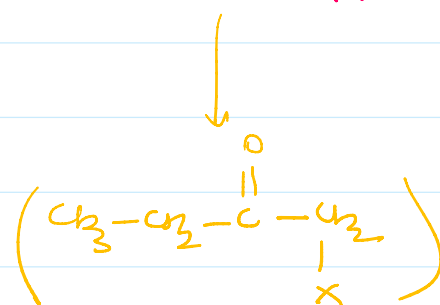
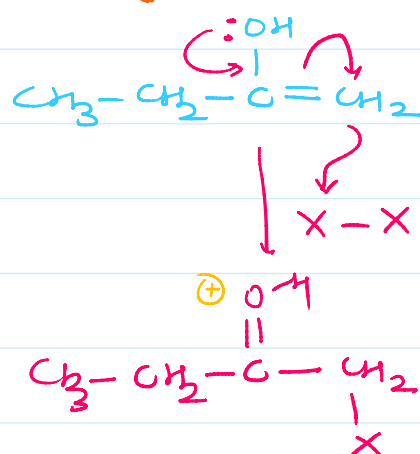
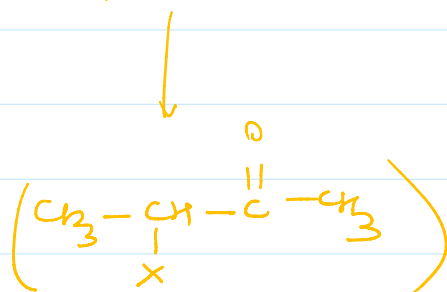
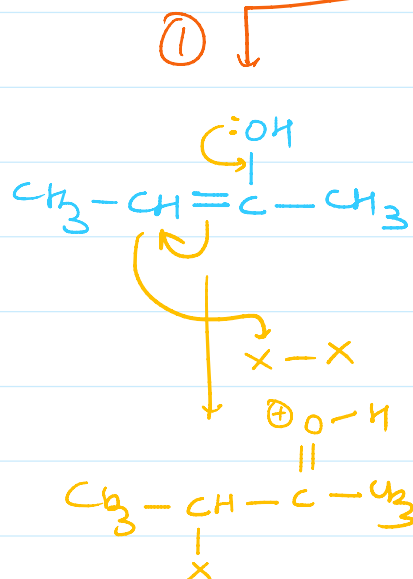


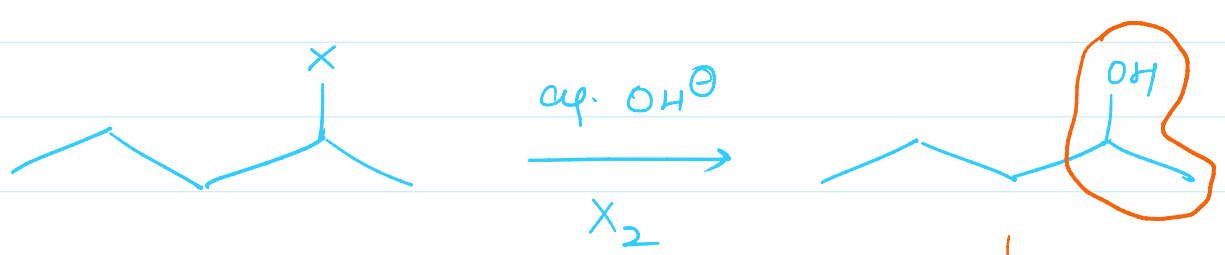
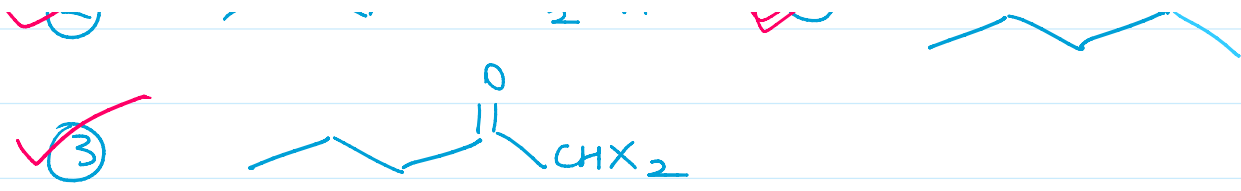
Haloform Reaction in acidic Medium:-



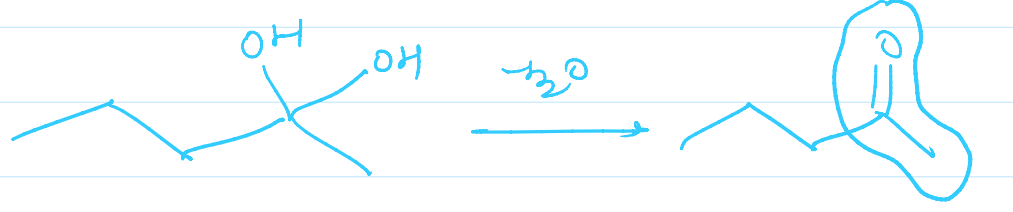
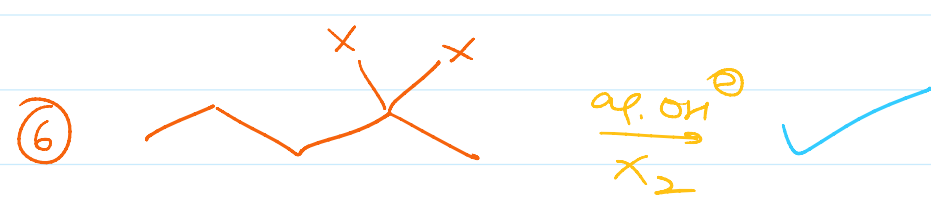


* acidic strength $\text{H}_1 < \text{H}_2$

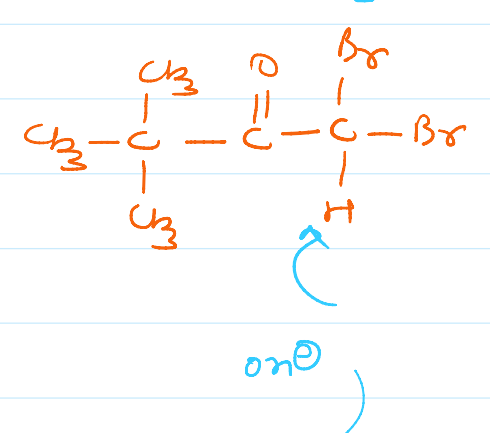
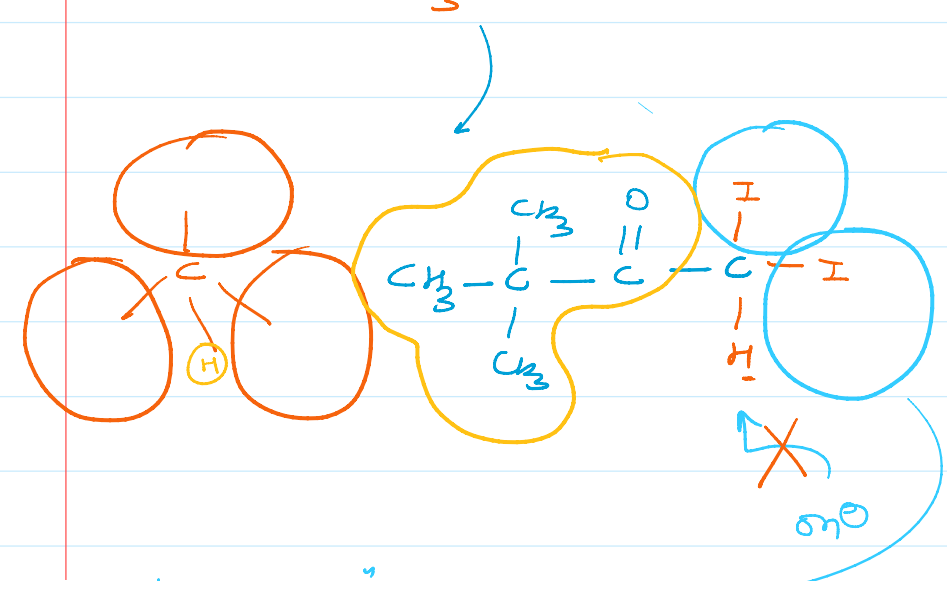
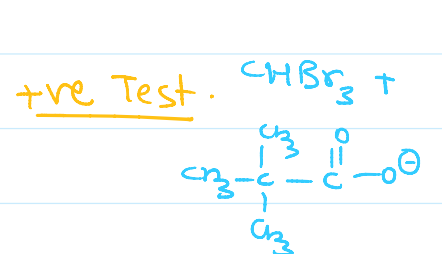
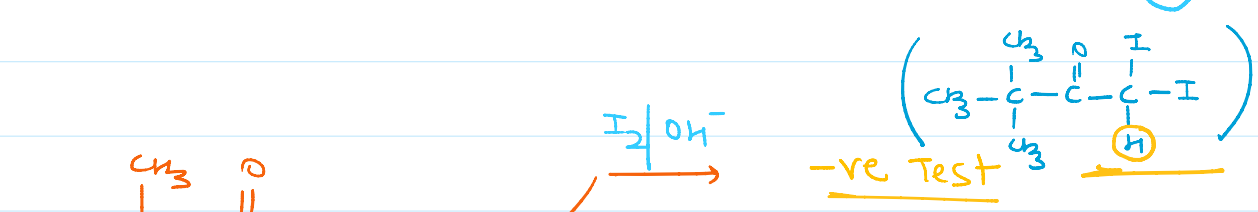




✓ +ve Test.

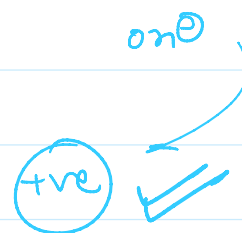


Bonus Q.

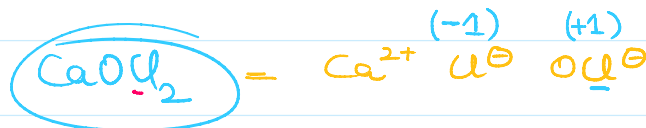
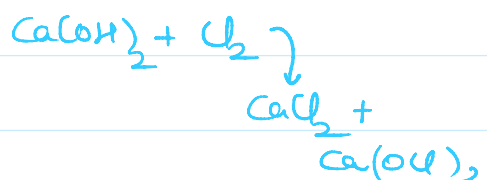
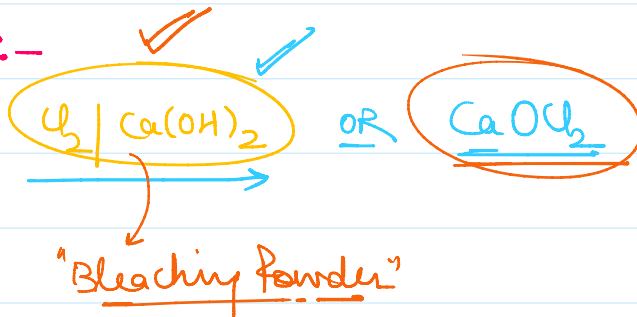
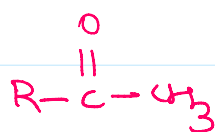


= 'Steric hindrance'

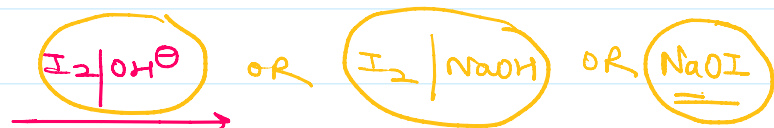
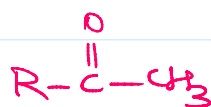
OH^- can not abstract H^+ ;



⊛ Chloroform Rxn:-



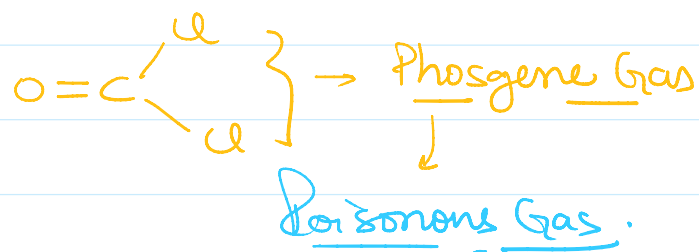
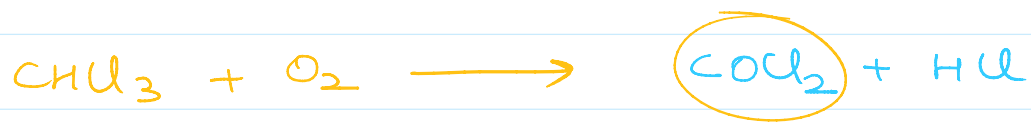
⊛⊛⊛ Iodoform!



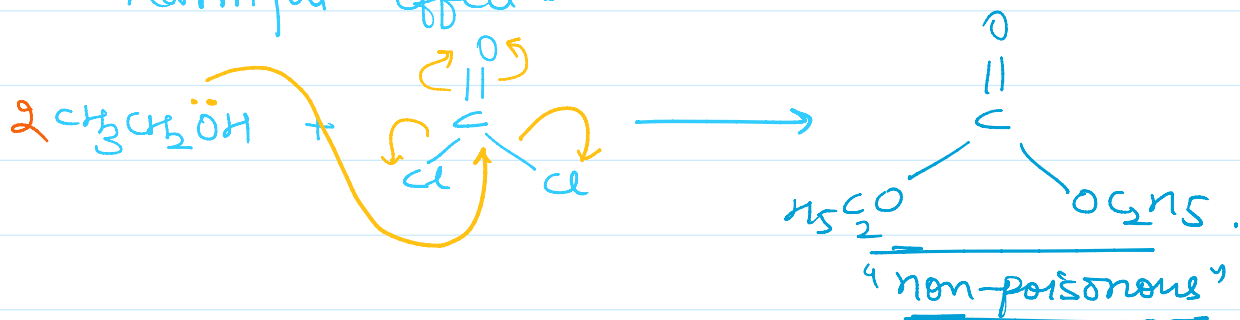
* Properties of Chloroform:-

liquid, was used as anaesthetic drug.

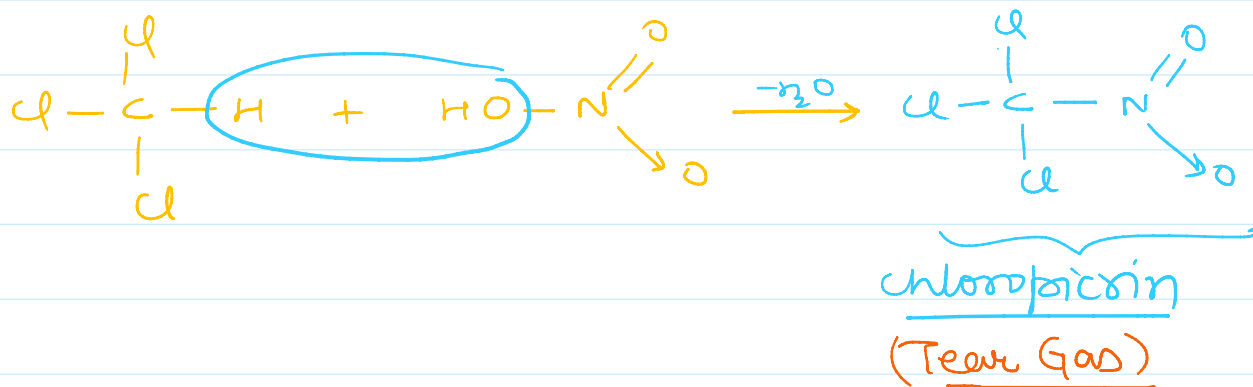
① Oxidation:-



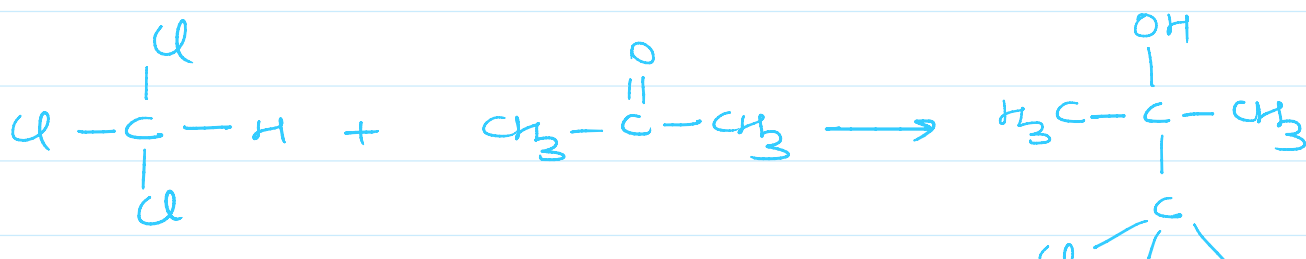
Ethanol is added in chloroform to protect from its harmful effect.

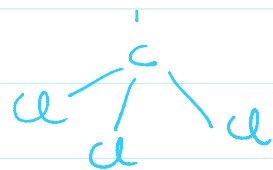


② Rxn with HNO_3 :-



③; Rxn with acetone:-

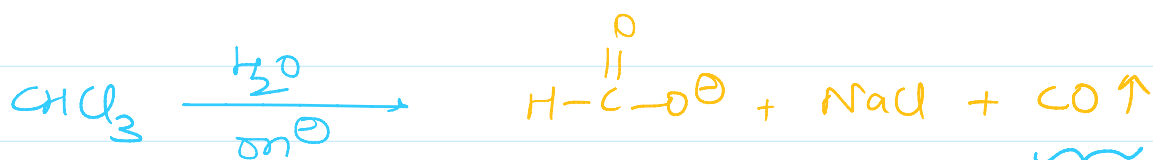




← "Chloretone"

Used as Sleep inducing agent

④ Hydrolysis of CHCl_3 :-



Best Method to prepare:

R-F = Swartz Rxn

R-Cl = Darzens Process

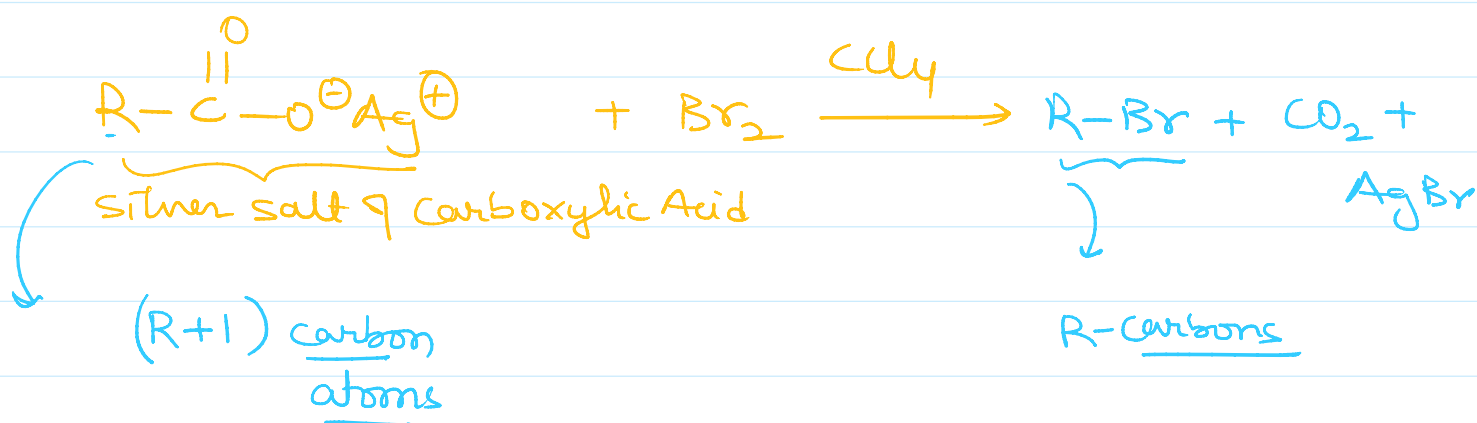
* R-Br = "Borodine Hunsdicker Rxn"

R-I = Finkelstein Rxn

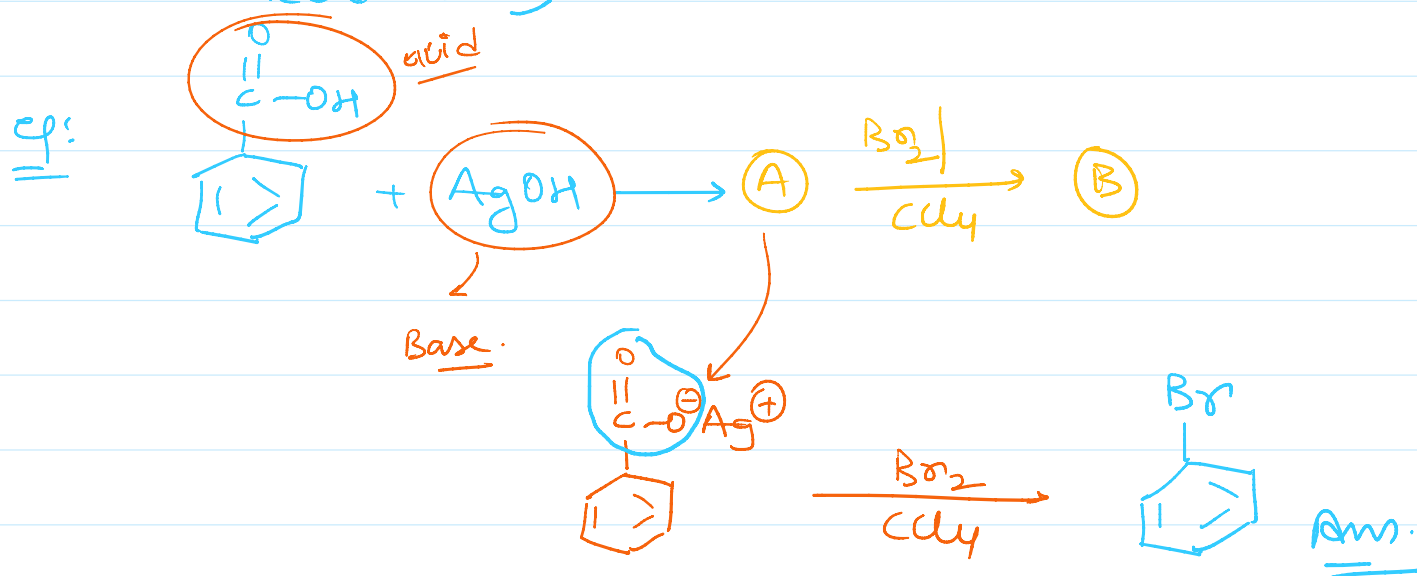
Borodine Hunsdicker Rxn
(Step down Rxn)

Silver salt of Carboxylic acid on rxn with Br_2 in CCl_4 gives alkyl bromide with

one carbon less.



- ① This rxn is a type of decarboxylation Rxn
- ② This rxn is a Step down Rxn i.e. Product is having 1-carbon less as compared to substrate
- ③ The rxn undergo via free Radical mechanism.



* Simonini Rxn:-

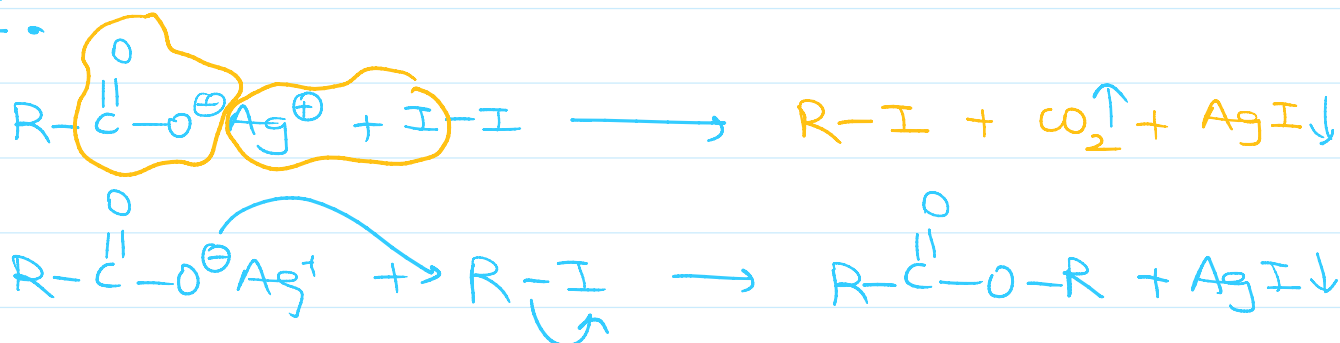
* Simonini Rxn:-

If in Boordine Hunsdicker Rxn, I_2 is used, it is called as Simonini Rxn.

In this Sam the major product is ester.



Mech:



Read NCERT

H.W

complete module
Alkyl & Aryl
Halide

* Redox Rxn in
Organic chemistry